Leak Diagnosis Checklist

A structured method for tracing a leak back to its true source — EDITABLE VERSION

Property / Site:

Inspector: **Date:**

Location of visible sign:

First noticed: **Weather since:**

Active during/after rain? Wind direction?

Water almost never shows up where it gets in. It travels along framing, insulation, and sheathing before gravity finally lets it drip somewhere visible. Chasing the stain instead of the water is the most common reason leak repairs fail on the first try. This checklist gives a repeatable method for working backward from the visible sign to the actual entry point.

1. The Golden Rule: Water Travels

Before inspecting anything, internalize this: water follows the path of least resistance downhill, along the top of whatever surface it lands on, often traveling several feet horizontally before finding a gap to drip through. A ceiling stain is a drip point, not necessarily an entry point.

- The entry point is almost always higher, and often to one side of, the visible stain
- Water travels along the underside of roof decking and down the face of framing members until it meets an obstruction
- Insulation acts like a wick and will move water sideways further than expected
- More than one leak source can feed the same stain - fixing the obvious one doesn't guarantee the stain stops

2. Step-by-Step Diagnostic Sequence

Work in this order. Do not patch or seal anything until the source is confirmed - sealing the wrong spot just hides the problem until it reappears somewhere new.

1 **Document the Visible Sign**

Photograph the stain, drip point, or damage with a scale reference. Note shape, size, and whether it's active, dried, or has multiple rings (indicating repeat events).

2 **Establish the Timing Pattern**

Ask when it appears: during rain, after rain, only in wind from a specific direction, or with no correlation to weather at all (which points to plumbing or condensation, not the exterior).

3 **Inspect the Attic or Cavity Above**

Before touching the roof or exterior, check the attic, ceiling cavity, or wall cavity directly above and around the stain for wet insulation, staining on framing, rust on fasteners, or mineral deposits - this narrows the search area significantly.

4 **Trace Upslope Along Framing**

Follow the wet framing member or sheathing seam upward and outward from the stain until the trail goes dry. The point where staining stops is close to the true entry point.

5 **Cross-Reference the Exterior**

Once a candidate area is identified from the inside, inspect the corresponding exterior location: roof penetrations, flashing, siding laps, window/door perimeters, or wall penetrations in that zone.

6 **Rule Out Non-Roof Sources**

Check for plumbing lines, HVAC condensate lines, or condensation on cold surfaces in the same cavity before assuming the roof or exterior wall is at fault - these mimic exterior leaks closely.

7 **Perform a Controlled Water Test**

If the source isn't visually obvious, run the hose test protocol (Section 4) to reproduce the leak under controlled, observed conditions.

8 **Confirm with a Second Data Point**

Before repairing, confirm the same source with at least two independent observations (e.g., visual trace + successful hose test) to avoid chasing a false lead.

9 **Document Findings Before Repair**

Photograph the confirmed entry point and record it against the original stain location - this becomes the baseline if the leak recurs.

3. Symptom-to-Source Map

Use this as a starting hypothesis only - always confirm with tracing or a water test before repairing.

VISIBLE SYMPTOM	LIKELY TRUE SOURCE
Ceiling stain near an exterior wall, upper floor	Roof flashing, siding lap, or wall penetration directly above/behind that wall section
Stain in the middle of a ceiling under a flat/low-slope roof	Membrane seam, penetration, or ponding area on the roof directly above - check a wider radius than the stain suggests
Staining or damage around a recessed light or ceiling fan	Attic penetration boot or fixture housing where insulation/vapor barrier was cut
Stain that appears only during wind-driven rain from one direction	Siding lap, window flashing, or wall penetration on the windward-facing elevation
Musty smell, no visible stain, or intermittent damp patch	Slow leak inside a wall cavity, or condensation on a poorly insulated cold surface (duct, pipe)
Stain at the top of an interior wall where it meets the ceiling	Wall-to-roof transition, parapet flashing, or kick-out flashing failure at that wall line
Stain below a bathroom or kitchen, no roof above	Plumbing supply/drain line, shower pan, or fixture seal - not a building-envelope leak
Water appears near a chimney or skylight only in heavy rain	Cricket/saddle flashing or counterflashing at the upslope side of the penetration

4. Controlled Water (Hose) Test Protocol

A hose test reproduces the leak under controlled conditions so you can watch, in real time, exactly where water enters. Two people are required: one applying water outside, one observing inside with a flashlight and phone for communication.

1. **Start Low, Not at the Suspected Source**
   Begin at the lowest point of the suspect area (e.g., base of a wall or bottom of a flashing run), not directly at your best guess. This confirms whether water below the guess is already a problem.

2. **Isolate One Component at a Time**
   Wet only one section - one window, one flashing run, one siding course - for 10-15 minutes before moving on. Never spray the whole elevation at once.

3. **Hold Position, Don't Sweep**
   Keep the hose in a steady, low-pressure stream simulating rain, not a sweeping or high-pressure spray that can force water past details that would never leak in real rain.

4. **Wait Before Concluding a 'Pass'**
   Absence of a leak in 10-15 minutes doesn't rule out that component. Some paths take longer. If time allows, extend to 20-30 minutes for suspect areas.

5. **Move Upward and Outward Systematically**
   If no leak appears, move to the next component up or beside the previous one, following the same direction water would naturally travel.

6. **Log Time-to-Appearance**
   Record how many minutes elapse between starting water at a given spot and the leak appearing inside. This elapsed time is useful if the leak needs to be reproduced again later.

7. **Stop as Soon as the Leak Reproduces**
   Once confirmed, stop the test immediately, mark the exact exterior spot, and photograph both the exterior mark and the interior drip simultaneously if possible.

5. On-Site Inspection Checklist

Work through each zone relevant to the suspected leak path. Mark OK / Issue / N-A and note details for anything flagged.

Interior / Attic

ITEM	OK	Issue	N/A	NOTES
Stain traced to a specific framing member or sheathing seam	☐	☐	☐	
Insulation in the affected area is wet, compressed, or stained	☐	☐	☐	
No active plumbing or HVAC line in the same cavity	☐	☐	☐	
No condensation source (cold duct, uninsulated pipe) in the cavity	☐	☐	☐	
Multiple stain rings noted (indicates a recurring, not one-time, leak)	☐	☐	☐	

Roof / Upper Exterior

ITEM	OK	Issue	N/A	NOTES
Flashing intact at the point directly above the traced path	☐	☐	☐	
No membrane seam separation, blistering, or puncture in that zone	☐	☐	☐	
No ponding water sitting above the suspected entry point	☐	☐	☐	
Penetration boots/seals intact in the traced zone	☐	☐	☐	

Walls / Siding / Windows

ITEM	OK	Issue	N/A	NOTES
Siding laps intact and properly overlapped in the traced zone	☐	☐	☐	
Window/door flashing and sealant intact at the traced elevation	☐	☐	☐	
No cracked or missing sealant at trim or penetration points	☐	☐	☐	
Kick-out flashing present where a roof plane meets a wall	☐	☐	☐	

Plumbing / Mechanical (if ruled in)

ITEM	OK	Issue	N/A	NOTES
No visible leak at supply or drain lines in the cavity	☐	☐	☐	
HVAC condensate line and pan draining correctly, no overflow	☐	☐	☐	
Fixture seals (tub, shower, sink) intact with no gaps	☐	☐	☐	

Before you repair: confirm the source with at least two independent signs (visual trace + hose test, or two separate hose tests on different days). A single observation is a hypothesis, not a diagnosis.

Confirmed Source & Notes

Inspector Signature: _____ **Date:** _____

This checklist is a general field aid for methodical leak tracing and does not replace evaluation by a licensed roofing, waterproofing, or plumbing professional for complex or recurring leaks.